

SAT Math Mastery · Week TP1C**Trick Pack 1 — Part C — Answer Key & Worked Solutions****Time** 120 minutes: 90 min solving + 30 min self-check and error log**Topics** exponent rewriting · mixture balance · parallel sides in polygons · similar triangles · discriminant conditions · composite functions · coefficient parity · quadratic modeling · trig identities · exponential ratios**Name / Date** _____**1 · SKILLS & IDEAS**

- S1.** Factor composite bases into primes, combine like powers, and shift an exponent so a chosen function value becomes the coefficient.
- S2.** Track one component through a multi-step mixing process with a single balance equation.
- S3.** Use parallel sides (supplementary co-interior angles) with the polygon angle sum, and angle bisectors in a trapezoid.
- S4.** Use similar triangles together with $(a + b)^2 = a^2 + b^2 + 2ab$ to avoid solving a quadratic.
- S5.** Turn "exactly one solution", "tangent", or "one x -intercept" into discriminant = 0, then apply the stated restriction.
- S6.** Build quadratics from vertex, roots, or data; evaluate composites by substituting the target input first.
- S7.** Reciprocal pairs b^x, b^{-x} and positivity of exponentials; constant-ratio exponentials and percent change.
- S8.** Prove or refute "must be" claims about coefficients with exhaustive parity or divisibility cases.
- S9.** Complementary angles: $\sin(90^\circ - \theta) = \cos \theta$, and sums versus products of ratios.

2 · REQUIRED FACTS

- $b^x > 0$ for every real x when $b > 0$; $b^{-x} = \frac{1}{b^x}$; $b^{x+k} = b^k \cdot b^x$.
- $ax^2 + bx + c = 0$ has exactly one real solution $\iff b^2 - 4ac = 0$; a line is tangent to a parabola $\iff$ the equated quadratic has discriminant 0.
- Co-interior (same-side) angles between parallel lines sum to 180° ; interior angles of an n -gon sum to $(n-2) \cdot 180^\circ$.
- Angles in a triangle sum to 180° ; a line parallel to a side of a triangle cuts off a similar triangle.
- If $\alpha + \beta = 90^\circ$ then $\sin \alpha = \cos \beta$ and $\tan \alpha = \frac{1}{\tan \beta}$; $\sin^2 \theta + \cos^2 \theta = 1$.
- A percent decrease of $p\%$ per step means a constant ratio $b = 1 - \frac{p}{100}$.

3 · REQUIRED FORMULAS

$$f(x) = a(x - h)^2 + k, \quad f(x) = a(x - r_1)(x - r_2), \quad \Delta = b^2 - 4ac, \quad (u + v)^2 = u^2 + v^2 + 2uv$$

$$\text{amount} = \text{fraction} \times \text{mass}, \quad \tan \theta + \cot \theta = \frac{1}{\sin \theta \cos \theta}, \quad f(x) = f(x_0) b^{x-x_0}$$

4 · MINI THEORY SHEET

Rewrite, then read. Factor every base into primes first; collect each prime's exponent; a single coefficient times a single base is the target form.

Parameter conditions. *One solution / tangent / one x -intercept* $\Rightarrow \Delta = 0$. Solve the resulting equation in the parameter and then use the sign or size restriction stated in the question.

Composite functions. Never expand. Compute the inner value at the target input, then feed it to the outer function.

Speed shortcuts.

Situation	Shortcut
composite bases in a power	prime-factor every base; one coefficient times one base
want $f(x_0)$ as the coefficient	$f(x) = f(x_0) b^{x-x_0}$
mixing with added pure component	one balance: original + added = fraction $\times$ final total
parallel sides in a polygon	co-interior angles sum to 180° ; then use $(n-2)180^\circ$
inscribed square in a right triangle	$s = \frac{ab}{a+b}$ and $(a+b)^2 = c^2 + 4 \cdot \text{Area}$
one real root / tangent	$\Delta = 0$; the sign condition picks the root
coefficient from two points	add and subtract equations to isolate b and $a+c$
vertex plus one point	$a(x-h)^2 + k$; composite: inner first
b^p and b^{-p} both given	set $t = b^p$, subtract, keep $t > 0$
"must be true" about integer coefficients	list all factor pairs; check parity or remainder in each
$\sin \theta + \cos \theta$ given	square it for $\sin \theta \cos \theta$
constant percent change	divide values to cancel the coefficient; shift is irrelevant

5 · GUIDED EXAMPLES**Example 1 — Shift the exponent to expose a value**

$f(x) = \frac{4^x \cdot 6^{x+1}}{9^x}$. Write f so that $f(3)$ is the coefficient.

$$4^x = 2^{2x}, 6^{x+1} = 2^{x+1} 3^{x+1}, 9^x = 3^{2x}, \text{ so } f(x) = 2^{3x+1} 3^{1-x} = 6 \cdot \left(\frac{8}{3}\right)^x.$$

$$f(3) = 6 \cdot \frac{512}{27} = \frac{1024}{9}, \text{ hence } f(x) = \frac{1024}{9} \left(\frac{8}{3}\right)^{x-3}.$$

Example 2 — Tangent line with a parameter

For which positive k is $y = kx - 1$ tangent to $y = x^2 + 3x + 3$?

$$x^2 + 3x + 3 = kx - 1 \Rightarrow x^2 + (3-k)x + 4 = 0. \Delta = (3-k)^2 - 16 = 0 \Rightarrow k = 7 \text{ or } k = -1. \text{ Positive: } k = 7.$$

Example 3 — A sum gives a product, then a reciprocal

For acute θ , $\sin \theta + \cos \theta = \frac{\sqrt{7}}{2}$. Find $\tan \theta + \tan(90^\circ - \theta)$.

$$\text{Square: } 1 + 2 \sin \theta \cos \theta = \frac{7}{4} \Rightarrow \sin \theta \cos \theta = \frac{3}{8}. \text{ Then } \tan \theta + \cot \theta = \frac{1}{\sin \theta \cos \theta} = \frac{8}{3}.$$

6 · PRACTICE SET — 15 SAT CHALLENGE PROBLEMS

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1.

SAT CHALLENGE · CALCULATOR

The function f is defined by $f(x) = \frac{9^x \cdot 12^{x+1}}{8^x}$. Which of the following expressions is equivalent to $f(x)$ and displays the value of $f(2)$ as a coefficient?

(A) $\frac{729}{4} \left(\frac{27}{2}\right)^{x-2}$

(B) $2187 \left(\frac{27}{2}\right)^{x-2} \checkmark$

(C) $2187 \left(\frac{27}{2}\right)^{x+2}$

(D) $162 \left(\frac{27}{2}\right)^{x-1}$

Worked solution

1. Prime factors: $9^x = 3^{2x}$, $12^{x+1} = 2^{2x+2} \cdot 3^{x+1}$, $8^x = 2^{3x}$.

2. $f(x) = 3^{2x+x+1} \cdot 2^{2x+2-3x} = 3^{3x+1} \cdot 2^{2-x} = 12 \left(\frac{27}{2}\right)^x$.

3. $f(2) = 12 \cdot \frac{729}{4} = 2187$.

4. Shift the exponent: $f(x) = f(2) \left(\frac{27}{2}\right)^{x-2} = 2187 \left(\frac{27}{2}\right)^{x-2}$.

5. Other options: $162 \left(\frac{27}{2}\right)^{x-1}$ is equivalent but shows $f(1)$; $\frac{729}{4} \left(\frac{27}{2}\right)^{x-2}$ and the $x+2$ form are not equivalent.

Answer: (B)

2.

SAT CHALLENGE · CALCULATOR

A metalworker melts together 150 g of alloy X, which is 20% nickel by mass, and m g of alloy Y, which is 50% nickel by mass. Then 40 g of pure nickel is stirred into the melt. The final alloy is 40% nickel by mass. What is the value of m ?

(A) 250

(B) 460

(C) 60 $\checkmark$

(D) 100

Worked solution

1. Final mass: $150 + m + 40 = 190 + m$, and nickel is 40% of it: $0.4(190 + m)$.

2. Nickel in: $0.20(150) + 0.50m + 40 = 70 + 0.5m$.

3. Balance: $70 + 0.5m = 0.4(190 + m) \Rightarrow 0.1m = 6 \Rightarrow m = 60$.

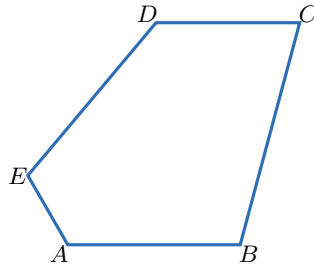
4. Check: total 250 g, nickel $30 + 30 + 40 = 100$ g, and $100/250 = 40\%$.

Answer: (C)

3.

SAT CHALLENGE · NO CALCULATOR NEEDED

In convex pentagon $ABCDE$ shown (not drawn to scale), side $\overline{AB}$ is parallel to side $\overline{DC}$. The measures of four interior angles are $\angle A = (6x)^\circ$, $\angle B = (5x + 5)^\circ$, $\angle C = (3x + 15)^\circ$ and $\angle D = (7x - 10)^\circ$. What is the measure, in degrees, of $\angle E$?



(A) 110 ✓

(B) 70

(C) 105

(D) 130

Worked solution

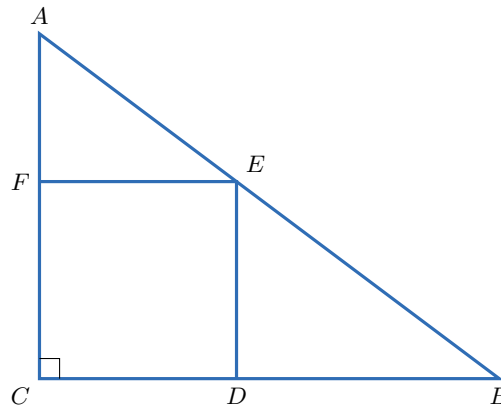
1. $\overline{AB} \parallel \overline{DC}$ with transversal $\overline{BC}$: $\angle B + \angle C = 180^\circ$ (co-interior).
2. $(5x + 5) + (3x + 15) = 180 \Rightarrow 8x = 160 \Rightarrow x = 20$.
3. $\angle A = 120^\circ$, $\angle B = 105^\circ$, $\angle C = 75^\circ$, $\angle D = 130^\circ$.
4. Pentagon angle sum $= (5 - 2) \cdot 180^\circ = 540^\circ$.
5. $\angle E = 540 - (120 + 105 + 75 + 130) = 110^\circ$.

Answer: (A)

4.

SAT CHALLENGE · NO CALCULATOR NEEDED

In right triangle ABC , $\angle C = 90^\circ$, $AB = 20$, and the area of the triangle is 96. Square $CDEF$ is inscribed in the triangle, with D on $\overline{BC}$, F on $\overline{AC}$, and E on $\overline{AB}$. What is the side length of the square?



(A) $\frac{24}{7}$

(B) $\frac{48}{5}$

(C) $\frac{96}{7}$

(D) $\frac{48}{7}$ ✓

Worked solution

1. Let the legs be $a = BC$, $b = AC$. Area: $\frac{1}{2}ab = 96 \Rightarrow ab = 192$. Pythagorean: $a^2 + b^2 = 400$.

2. $(a + b)^2 = a^2 + b^2 + 2ab = 400 + 384 = 784 \Rightarrow a + b = 28$ (no quadratic needed).

3. Triangle $FAE \sim$ triangle CAB (since $\overline{FE} \parallel \overline{CB}$): $\frac{AF}{AC} = \frac{FE}{CB}$, i.e. $\frac{b-s}{b} = \frac{s}{a}$.

4. This gives $s \left(\frac{1}{a} + \frac{1}{b} \right) = 1$, so $s = \frac{ab}{a+b} = \frac{192}{28} = \frac{48}{7}$.

5. (Legs are 12 and 16; $\frac{12 \cdot 16}{28} = \frac{48}{7}$.)

Answer: (D)

5.

SAT CHALLENGE · NO CALCULATOR NEEDED

The function f is defined by $f(x) = ax^2 + bx + c$, where a , b , and c are constants. The graph of $y = f(x)$ in the xy -plane passes through $(1, 8)$ and $(-1, 2)$ and has exactly one x -intercept. What is the greatest possible value of a ?

(A) $\frac{5}{2}$

(B) 9

(C) $\frac{9}{2}$ ✓

(D) $\frac{1}{2}$

Worked solution

1. $f(1) = a + b + c = 8$ and $f(-1) = a - b + c = 2$. Subtract: $2b = 6 \Rightarrow b = 3$. Add: $a + c = 5 \Rightarrow c = 5 - a$.

2. Exactly one x -intercept: $b^2 - 4ac = 0 \Rightarrow 9 = 4a(5 - a)$.

3. $4a^2 - 20a + 9 = 0 \Rightarrow (2a - 1)(2a - 9) = 0 \Rightarrow a = \frac{1}{2}$ or $a = \frac{9}{2}$.

4. The greatest possible value is $\frac{9}{2}$.

Answer: (C)

6.

SAT CHALLENGE · NO CALCULATOR NEEDED

The quadratic function f has its vertex at $(3, -2)$, and $f(5) = 6$. The function g is defined by $g(x) = \frac{3}{4}f\left(\frac{1}{2}x - 1\right)$. The graph of $y = f(x)$ has y -intercept $(0, p)$ and the graph of $y = g(x)$ has y -intercept $(0, q)$. What is the value of $q - p$?

(A) $-\frac{23}{2}$

(B) $\frac{13}{2}$ ✓

(C) -4

(D) $\frac{45}{2}$

Worked solution

1. Vertex form: $f(x) = a(x - 3)^2 - 2$. From $f(5) = 6$: $4a - 2 = 6 \Rightarrow a = 2$, so $f(x) = 2(x - 3)^2 - 2$.

2. $p = f(0) = 2(9) - 2 = 16$.

3. $q = g(0) = \frac{3}{4}f\left(\frac{1}{2} \cdot 0 - 1\right) = \frac{3}{4}f(-1)$, and $f(-1) = 2(16) - 2 = 30$, so $q = \frac{45}{2}$.

4. $q - p = \frac{45}{2} - 16 = \frac{13}{2}$.

Answer: (B)

7.

SAT CHALLENGE · NO CALCULATOR NEEDED

The function f is defined by $f(x) = 2^x + k$, where k is a constant. For a real number p , $f(p) = 9$ and $f(-p) = \frac{21}{4}$. What is the value of p ?

Worked solution

1. Let $t = 2^p$, so $t > 0$ and $2^{-p} = \frac{1}{t}$. Then $t + k = 9$ and $\frac{1}{t} + k = \frac{21}{4}$.

2. Subtract to eliminate k : $t - \frac{1}{t} = \frac{15}{4} \Rightarrow 4t^2 - 15t - 4 = 0 \Rightarrow (4t + 1)(t - 4) = 0$.

3. $t = -\frac{1}{4}$ is rejected because $2^p > 0$; so $t = 4$.

4. $2^p = 4 \Rightarrow p = 2$. (Check: $k = 5$; $f(2) = 9$, $f(-2) = \frac{1}{4} + 5 = \frac{21}{4}$.)

Answer: 2

8.

SAT CHALLENGE · NO CALCULATOR NEEDED

For integers a , b , c , and d , $(ax + b)(cx + d) = 6x^2 + kx + 5$ for all values of x , where k is a constant. Which of the following must be true?

I. k is oddII. When k is divided by 4, the remainder is 1III. k is not divisible by 5

(A) II and III only

(B) I, II, and III

(C) I only

(D) I and III only ✓

Worked solution

1. Matching coefficients: $ac = 6$, $bd = 5$, $k = ad + bc$.

2. Since $bd = 5$ is odd, b and d are both odd. Since $ac = 6$ is not divisible by 4, exactly one of a, c is even. Then $k = ad + bc \equiv a + c \equiv 1 \pmod{2}$. So I is true.

3. Since $bd = 5$, exactly one of b, d is a multiple of 5; say $5 \mid b$. Then $k \equiv ad \pmod{5}$, and $a \mid 6$ and $d = \pm 1$ are not multiples of 5. (Same if $5 \mid d$.) So III is true.

4. II: $(a, c, b, d) = (1, 6, 5, 1)$ gives $k = 1 \cdot 1 + 5 \cdot 6 = 31$, remainder 3 on division by 4. So II is false, even though $k = 13$ and $k = 17$ satisfy it.

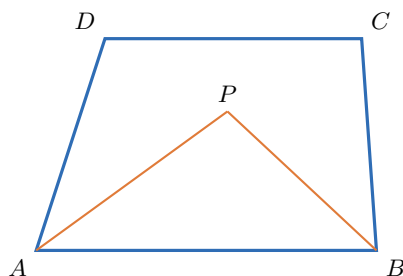
5. Possible values: $k = \pm 11, \pm 13, \pm 17, \pm 31$. Only I and III must be true.

Answer: (D)

9.

SAT CHALLENGE · NO CALCULATOR NEEDED

In trapezoid $ABCD$ shown (not drawn to scale), $\overline{AB} \parallel \overline{DC}$, $\angle D = 108^\circ$, and $\angle C = 94^\circ$. The bisector of $\angle DAB$ and the bisector of $\angle ABC$ intersect at point P inside the trapezoid. What is the measure, in degrees, of the obtuse angle formed at P by these two bisectors?



(A) 101 ✓

(B) 79

(C) 22

(D) 108

Worked solution

- $\overline{AB} \parallel \overline{DC}$: $\angle A + \angle D = 180^\circ \Rightarrow \angle A = 72^\circ$; $\angle B + \angle C = 180^\circ \Rightarrow \angle B = 86^\circ$.
- Half-angles: $\angle PAB = 36^\circ$, $\angle PBA = 43^\circ$.
- In triangle APB : $\angle APB = 180 - 36 - 43 = 101^\circ$, which is obtuse.
- The vertical/supplementary angle is 79° (acute), so the obtuse angle is 101° .

Answer: (A)

10.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the xy -plane, the line $y = k(x - 1) - 2$, where k is a positive constant, is tangent to the parabola $y = x^2 + 2x + 4$. What is the value of k ?

(A) 8

(B) 10 ✓

(C) -2

(D) 4

Worked solution

- Set equal: $x^2 + 2x + 4 = kx - k - 2 \Rightarrow x^2 + (2 - k)x + (k + 6) = 0$.
- Tangency means exactly one solution: $(2 - k)^2 - 4(k + 6) = 0$.
- $k^2 - 8k - 20 = 0 \Rightarrow (k - 10)(k + 2) = 0 \Rightarrow k = 10$ or $k = -2$.
- Since $k > 0$, $k = 10$. (The tangent point is $x = \frac{k - 2}{2} = 4$, which is not the answer.)

Answer: (B)

11.

SAT CHALLENGE · CALCULATOR

The altitude h , in meters, of a small rocket t seconds after launch is modeled by $h(t) = at^2 + bt + c$, where a , b , and c are constants. At launch the altitude is 3 meters; 2 seconds after launch it is 43 meters; and 5 seconds after launch it is 28 meters. According to the model, what is the altitude, in meters, 3 seconds after launch?

Worked solution

- Launch altitude: $h(0) = c = 3$.
- $h(2) = 4a + 2b + 3 = 43 \Rightarrow 2a + b = 20$ (divide by 2).
- $h(5) = 25a + 5b + 3 = 28 \Rightarrow 5a + b = 5$ (divide by 5).
- Subtract: $3a = -15 \Rightarrow a = -5$; then $b = 20 - 2(-5) = 30$.
- $h(3) = -5(9) + 30(3) + 3 = -45 + 90 + 3 = 48$.

Answer: 48

12.

SAT CHALLENGE · NO CALCULATOR NEEDED

The quadratic function f has zeros at $x = -2$ and $x = 5$, and $f(1) = -24$. The function h is defined by $h(x) = f(x+2) - 3f(-x)$. What is the value of $h(1)$?

Worked solution

1. $f(x) = a(x+2)(x-5)$; $f(1) = a(3)(-4) = -24 \Rightarrow a = 2$, so $f(x) = 2(x+2)(x-5)$.
2. Substitute $x = 1$ first: $h(1) = f(3) - 3f(-1)$.
3. $f(3) = 2(5)(-2) = -20$ and $f(-1) = 2(1)(-6) = -12$.
4. $h(1) = -20 - 3(-12) = -20 + 36 = 16$.

Answer: 16

13.

SAT CHALLENGE · NO CALCULATOR NEEDED

In right triangle PQR , $\angle R = 90^\circ$ and $\sin P + \sin Q = \frac{\sqrt{6}}{2}$. What is the value of $\tan P + \tan Q$?

Worked solution

1. $\angle Q = 90^\circ - \angle P$, so $\sin Q = \cos P$ and $\tan Q = \frac{1}{\tan P}$.
2. Square the given sum: $\sin^2 P + 2 \sin P \cos P + \cos^2 P = \frac{6}{4} \Rightarrow 1 + 2 \sin P \cos P = \frac{3}{2}$.
3. $\sin P \cos P = \frac{1}{4}$.
4. $\tan P + \tan Q = \frac{\sin P}{\cos P} + \frac{\cos P}{\sin P} = \frac{1}{\sin P \cos P} = 4$.

Answer: 4

14.

SAT CHALLENGE · NO CALCULATOR NEEDED

The quadratic function f has an axis of symmetry at $x = 2$, and its graph passes through $(3, -3)$ and $(5, 13)$. The functions g and h are defined by $g(x) = \frac{x-6}{2}$ and $h(x) = 3 - f(g(x))$. What is the y -coordinate of the y -intercept of the graph of $y = h(x)$?

Worked solution

1. Vertex form with axis $x = 2$: $f(x) = a(x-2)^2 + k$.
2. $f(3) = a + k = -3$ and $f(5) = 9a + k = 13$. Subtract: $8a = 16 \Rightarrow a = 2$, $k = -5$.
3. $f(x) = 2(x-2)^2 - 5$. The y -intercept of h is $h(0)$.
4. Inner value first: $g(0) = -3$. Then $f(-3) = 2(-5)^2 - 5 = 45$.
5. $h(0) = 3 - 45 = -42$.

Answer: -42

15.

SAT CHALLENGE · NO CALCULATOR NEEDED

The function f is defined by $f(x) = c \cdot b^{x+3}$, where c and b are positive constants. If $f(1) = 1296$ and $f(3) = 729$, then for each increase of 1 in the value of x , $f(x)$ decreases by $p\%$. What is the value of p ?

Worked solution

1. $f(1) = cb^4 = 1296$ and $f(3) = cb^6 = 729$.
2. Divide to cancel c : $b^2 = \frac{729}{1296} = \frac{9}{16}$. Since $b > 0$, $b = \frac{3}{4}$.
3. Each unit increase in x multiplies f by $\frac{3}{4}$, a decrease of $1 - \frac{3}{4} = \frac{1}{4} = 25\%$.
4. $p = 25$. (The value of c is not needed.)

Answer: 25